



Review

Building Information Modeling and Digital Twins for Functional and Technical Design of Smart Buildings with Distributed IoT Networks—Review and New Challenges Discussion

Gabriela Walczyk and Andrzej Ożadowicz *

Department of Power Electronics and Energy Control Systems, Faculty of Electrical Engineering, Automatics, Computer Science and Biomedical Engineering, AGH University of Krakow, al. Mickiewicza 30, 30-059 Krakow, Poland; gabiwalczyk@agh.edu.pl

* Correspondence: ozadow@agh.edu.pl

Abstract: Modern building automation systems implement plenty of advanced control and monitoring functions that consider various parameters like users' activity, lighting, temperature changes, etc. Moreover, novel solutions based on the Internet of Things and cloud services are also being developed for smart buildings to ensure comfort of use, user safety, energy efficiency improvements, and integration with smart grids and smart city platforms. Such a wide spectrum of technologies and functions requires a novel approach in building automation systems design to provide effective implementation and flexibility during operation. At the same time, in the building design and operation industries, tools based on building information modeling and digital twins are being developed. This paper discusses the development directions and application areas of these solutions, identifying new trends and possibilities of their use in smart homes and buildings. In particular, the focus is on procedures for selecting automation functions, effective integration, and interoperability of building management systems with the Internet of Things, considering the organization of prediction mechanisms and dynamic functional changes in buildings and smart networks. Chosen solutions and functions should consider the requirements set out in the EN ISO 52120 standard and the guidelines defined for the Smart Readiness Indicator.



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Keywords: building information modeling; digital twin; building automation and control systems; Internet of Things; energy efficiency; EN ISO 52120; Smart Readiness Indicator

1. Introduction

The modern building construction industry is subject to numerous technological, functional, and qualitative changes. Particularly in Europe, the technical and functional requirements for new and modernized buildings are changing radically as a result of the publication of guidelines from various European Commission directives. One of the most important documents is the Energy Performance of Buildings Directive 2018 (EPBD 2018) [1], which determines the direction of changes and efforts to improve the energy performance of buildings. Its latest revision emphasizes the importance of building automation and control systems (BACSs) in these processes and introduces guidelines for estimating the Smart Readiness Indicator (SRI), which is intended to support building transformation processes. This is particularly important in the context of the expansion of the infrastructure associated with residential and nonresidential buildings. It integrates new devices and systems such as renewable energy sources (RES), electric vehicle charging stations, energy storage, and other elements of local energy microgrids. Moreover, the structures of information and communication technology (ICT) networks as well as fieldbus automation systems installed in buildings are evolving. The latest ICT of distributed Internet of Things (IoT) modules is also being implemented in various applications, supporting the comfort and convenience of using the increasingly pervasive infrastructure [2–4].

Therefore, it is necessary to change the approach in the design of buildings and their technical and utility infrastructure and also building management systems (BMS) using network solutions, IoT, and BACS, together with the implementation of energy management mechanisms and the enabling of the cooperation of buildings with intelligent power grids [5,6]. In this context, the main challenges can be summarized as follows: (i) the selection of BACS functions and the level of their integration in buildings, (ii) the estimation of the level of advancement of automation systems and smart solutions in buildings, and (iii) the estimation of the potential of smart systems and functions in buildings in terms of improving energy efficiency and their flexibility to support innovative demand-side management and demand-side response (DSM/DSR) functions [7–9].

1.1. Buildings Smartness and Energy Efficiency Evaluation

These core challenges regarding the selection and potential of functional concepts in smart buildings have been undertaken by engineering and research teams for several years [10–13], but new paths and solutions in this area are still being sought. The sources address answers relating to IoT technology, levels of automation, and the energy efficiency of buildings, mentioning both the residential and services sectors. Their results include, among others, the guidelines of the already mentioned EPBD directive, the EN 15232:2017 standard which was replaced by the new EN ISO 52120:2022 [14,15], and the latest SRI index initiative; these guidelines are still being developed, modified, and, most importantly, verified in case studies [16–18]. The guidelines from these documents classify the functions and services of smart systems in building applications, depending on their level of complexity and integration, and provide indicators that enable numerical determination of the level of improvement in energy efficiency and smartness of the applied BACS systems with IoT. Moreover, the EN ISO 52120 standard [15] and the SRI development report [19] define methodologies for calculating the mentioned indicators. It should be emphasized that in both cases, simplified and detailed methods are provided for carrying out the procedures for their estimation and calculation. Simplified methods are based on average indicators provided in the standard and report, which allow for a preliminary, rough estimation of the potential energy savings level or SRI percentage based on collected information about BACS and technical building management (TBM) functions and services, using classification lists of these functions and services specified in the standard and report as appropriate. In turn, in detailed methods, usually, multistage, average indicators are considered in the initial stage of calculations, but subsequent iterations require knowledge of additional, detailed data describing the specific use of a given building (utility profiles, technical data of specific building infrastructure subsystems, etc.). To provide more precise results, actual measurement data are preferred from an existing building in order to verify selected automation functions, proposed connections, integration, and parametric settings. Hence, it is of great importance to have case studies research conducted for various types of buildings located in different countries and parts of the world, which are the subject of discussion in scientific and technical publications [17,20–22]. They deal with buildings and measurements taken in both warm and cold climates, which compare the translation of theoretical assumptions into practice. The importance of quality and quantity of data on the methodology and result of SRI calculations has been proven among diverse experts, using the example of physical buildings in mentioned articles. It is rather obvious that detailed methods are more accurate and provide the possibility of more precise selection of BACS and TBM. However, they are time-consuming (several months) and require the organization of additional measurement functions, along with data collection and processing. Designers of architecture and technological installations of buildings have been struggling with similar problems for many years. To facilitate the work, they use innovative building information modeling (BIM) and digital twin (DT) technologies and tools [23–28]. Moreover, the advanced distributed control and monitoring networks with IoT nodes are considered as technological platforms supporting these solutions.

1.2. Review Approach and Methodology, Original Contribution, and Paper Structure

Bearing in mind these aspects, in this paper the authors provide an overview of BIM and DT methods with the most commonly used tools in the building industry. Areas of application, the main features that allow them to simplify and increase the accuracy and efficiency of the design, modeling and parametric analysis processes of various systems, installations, and complex building structures are analyzed. The selection of these issues was based on a preliminary review of publication topics and citation indicators of papers devoted to BIM and DT in the Web of Science and Scopus databases, as well as additional Google Scholar platform and patent databases (Google Patents, Espacenet, and Uspto search engines). In particular, the authors verified the number and type of publications in two of the most important databases of scientific and technical publications, which provide preliminary sorting of bibliographic data. The results are presented in Table 1.

Table 1. Results of the literature review in bibliometric databases.

Database	Publication Type	Search Term (for All of the Fields in Database)		
		BIM	DT	BIM + DT
Web of Science	Articles	111,477	10,433	642
	Review articles	5094	4672	96
Scopus	Articles	17,779	10,940	8388
	Review articles	1158	1084	2106

Furthermore, the number of publications over the years was verified in both databases. The results demonstrated that in the Web of Science database, publications with BIM topics have exceeded 1000 items per year since 2000 and those with DT topics since 2020, and the number of publications addressing both areas simultaneously has exceeded 100 per year since 2021. Similarly, for the Scopus database, publications with BIM topics have exceeded 1000 items per year since 2001 and with DT topics since 2019. And the number of publications addressing both areas simultaneously has exceeded 100 per year since 2017. In light of the aforementioned data, the authors decided to limit the years of analysis to those between 2010 and 2023, with a focus on journal and conference articles as well as reviews published during this period.

The results of the preliminary review were then subjected to a comprehensive review of the scientific and conference papers. It included publications from several bibliographic databases recognized in specific construction, electrical engineering, automation, and ICT industries. Consequently, the publication analysis for the purposes of this study was based on the databases of publishers whose publications were most frequently identified in the preliminary review, which were recognized in the technical sciences environment. These included Springer, ScienceDirect, MDPI, IEEE Xplore (journals and conferences), and additionally Taylor and Francis, the ACM Digital Library, and the Wiley Online Library. The findings of the review of these publication databases, with a focus on the period between 2010 and 2023 and engineering disciplines, are presented in Table 2.

Consequently, the final selection of publications were based on keywords related to the area of BIM and DT applications in building design, organization of control systems, and energy-efficiency improvement. Keywords were the most crucial element in the categorization of papers and publications, which were divided into several major thematic groups:

- BIM and DT (e.g., building information modeling, digital twin, BIM ontology, parallel system, deep reinforcement learning, machine learning, energy modeling, predictive maintenance, model-based system engineering, reinforcement learning, mathematical modeling, high level modeling, Industry Foundation Classes);
- Building automation (e.g., smart building, intelligent building, building automation, smart home, cyber-physical systems, resource management, facility management, Smart Readiness Indicator, building management);

- Distributed control systems (e.g., embedded systems, fieldbus networks, Internet of Things, mesh networks, edge and fog computing, multiaccess edge computing, fog-based architectures, frameworks, networking platform);
- Design (e.g., control systems design, service-oriented architecture, control as a service, servitization, automated control system design, network architecture and topology, smart buildings design, optimal sensor placement, smart retrofitting, sustainable building design, model-based design, digital twin design verification);
- Communication and data processing (e.g., semantic specification, semantic model, data model, ontology, data integration, reference data model, interoperability, workflow, cloud computing, communication standards, fieldbus standards, cyber security, data privacy);
- Energy management (e.g., energy management systems, home energy management system, load balancing and shifting, smart grid, microgrid, energy efficiency, energy flexibility, demand response, demand-side management, energy performance of buildings, EPBD).

Table 2. Results of the literature review in publishing databases.

Database	Publication Type	Search Term (for All of the Fields in Database)		
		BIM	DT	BIM + DT
Springer	Articles	14,394	2073	369
	Conference papers	11,631	1812	287
	Review articles	961	216	88
Science Direct Elsevier	Articles	531,416	26,288	5831
	Review articles	55,294	2677	1232
MDPI	Articles	4573	1587	130
	Review articles	289	259	27
IEEE Xplore	Articles	10,710	1297	104
	Conference papers	42,345	6343	479
Taylor and Francis	Articles	100,179	31,278	15,934
	Review articles	2499	761	479
ACM Digital Library	All types of publications	420,305	388,438	187,264
Wiley Online Library	Journal papers	962,144	297,491	129,909
	Books	114,702	38,876	22,847

Following the final selection and filtering of publications, a set of 75 items was identified as being relevant to the subject of building information modeling (BIM) and digital twins (DTs), as well as a range of detailed issues related to building automation, energy efficiency, and facility management. In this review, the majority of the selected publications were included, along with the additional literature pertaining to the technical and functional organization of BACS and BMS systems. Considering them, this paper makes a significant contribution to the field of BACS by examining the potential of using BIM and DT tools in the design and effective use of BACS with IoT elements in buildings. The authors identify three areas of original contribution and formulate the following theses:

1. The list of BACS functions defined in the EN ISO 52120 standard and the SRI service guidelines should be used as a framework for the selection and organization of key BIM functions supporting technical and functional BACS design. Furthermore, these guidelines can be used to optimize DT structures in buildings, particularly as a tool for dynamic and efficient energy management in buildings.
2. It is possible to use DT structures in the implementation of the detailed method of calculating SRI and the precise selection of BACS functions with the analysis of the

energy efficiency of buildings for use in the construction of buildings similar to those previously analyzed, and so on.

3. Technologies and solutions in the area of generic IoT and fieldbus networks (edge) can be employed as infrastructure for the implementation of DT functions during the operational phase of buildings with BACS and for the more precise selection of BIM model parameters for future building designs with a similar use and purpose.

The confrontation of these theses with the outcomes and conclusions presented in numerous publications related to the subject of the concept and practical implementation of BIM and DT techniques and technologies prompted the authors to prepare the review presented in this paper. In light of the preliminary analyses of the existing literature on the subject, in particular, numerous review publications on BIM and DT, it can be seen that the theses introduced by the authors clearly identify new potential application areas for these technologies and techniques, along with the gaps and challenges indicated by the literature analysis and discussed in this paper. Existing problems and a wide range of knowledge about existing solutions on the market, recorded in numerous mentioned publications, require systematization and wider discussion in the context of development directions, which the authors of the text undertake in this work. These factors determined the commencement of work on this review, which, according to the authors, introduces novel ideas and concepts of potential BIM and DT applications in the building automation industry and improving their energy efficiency.

The rest of this paper is organized as follows. Section 2 provides an overview of the current application areas of BIM and DT techniques and tools, accompanied by an analysis of the key challenges and gaps. Section 3 presents the latest trends and challenges associated with the design and maintenance of buildings using BIM and DT. In Section 4, the authors examine the potential of BIM and DT solutions to support BACS with IoT design, construction, and operation procedures in buildings. Finally, Section 5 presents the conclusions and outlines future work. Table 3 shows the most important abbreviations used in this paper and explains their meaning.

Table 3. Important abbreviations used to write this article that appear in the text.

Abbreviation	Extension
AI	Artificial intelligence
BaaS	Building as a service
BACS	Building automation and control system
BAS	Building automation system
BIM	Building information modeling
BMS	Building management system
CMMS	Computerized maintenance management system
DSM	Demand-side management
DSR	Demand-side response
DT	Digital twin
DTaaS	Digital twin as a service
FM	Facility management
HVAC	Heating, ventilation, air conditioning
ICT	Information and communications technology
IFCs	Industry Foundation Classes
IoT	Internet of Things
ML	Machine learning
RES	Renewable energy sources
SRI	Smart Readiness Indicator
TBM	Technical building management

2. BIM and DT Idea and Applications

For many years, the construction industry has been convinced that BIM is an approach and tool that primarily improves the design processes and effective implementation of mod-

ern buildings. The utilization of digital models of building structures, their installations, and their system infrastructure facilitate the collaboration of designers and contractors from various industries, thereby reducing the time required for completion of the project while simultaneously enhancing the quality of the final product [29,30]. Nevertheless, the technological advancements that have occurred over the past decade or so, encompassing the evolution of computer techniques, the development of advanced engineering software, and innovative technologies of building infrastructure systems, have led to the expansion of the areas of BIM applications also during the period of buildings' operation. At present, numerous industry experts and engineering teams emphasize that BIM can be used at any stage in the life cycle of a building or infrastructure, from design to operation, as indicated in both [30,31], summarized in Figure 1.

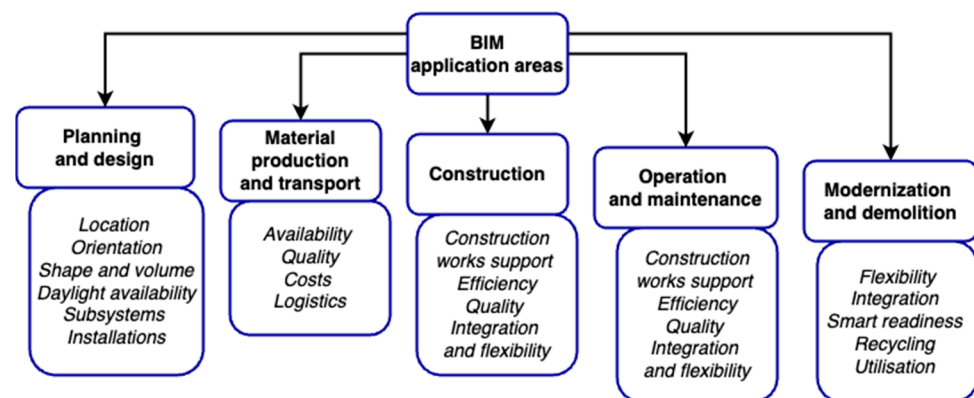


Figure 1. Contemporary, innovative areas of BIM applications.

The results of these research studies indicate that modern building design, and in particular smart building design, comprises the following stages where a BIM approach can be implemented:

1. The planning and design: This stage determines the purpose of the building, its functions, and how smart technologies can be integrated. At the design stage, the following aspects are typically considered in depth: a thorough analysis of the shape, geometric shape, and appearance of the building to assess the strength and stability of the building components. By designing heating, ventilation, air conditioning (HVAC), plumbing, and electrical systems, it ensures optimal use conditions and energy efficiency. The assessment of the environmental and energy impact of a building allows for the minimization of such impact. Considering building orientation, window–wall ratio and additional analyses, BIM allows engineers to create buildings that are not only functional and aesthetic but also sustainable and energy-efficient [31].
2. The production and transportation of material: The necessary building materials, including those specific to smart building systems, are manufactured and delivered to the site.
3. The construction: This stage involves the physical construction of the facility and the installation and integration of intelligent building technology and systems. The implementation of BIM at the construction stage encompasses monitoring of construction progress and occupational safety and health issues [32].
4. The operation and maintenance: Once completed, the building is used. This phase encompasses regular maintenance and maintenance of both the design itself and the intelligent systems. BIM after construction involves monitoring the functioning of a building, usually with a DT, and using the IoT with machine learning (ML) [33]. BIM is also used to assess the performance of buildings after construction, encompassing actual energy consumption as well as flexibility to dynamic changes [34].
5. Modernization and demolition: Over time, the building may require upgrading or incorporation of intelligent systems. Users and facility managers need to introduce

new technologies, and adapting to changing conditions, regulations, and technical and safety requirements is important [35]. Following a lengthy service life, demolition may be required considering the principles of disposal of building and construction materials and their potential for recycling [36].

This brief overview shows that multithreaded support for BIM tools is still developing. A particularly important aspect is the possibility of an integrated approach in the design and management of building infrastructure, which has been raised for many years by engineers and experts in the BACS building automation industry [6,37–39]. The concept of the universality of proposed functional blocks is not a new idea, but its application in the discussed subject is a challenge that has not yet been solved and is described rather than initiated. The ability to introduce integrity between the various parts of a building system, regardless of the software or device manufacturer, is an attractive vision of the future, discussed in the sources mentioned above. The development of the BIM implementation concept in the BACS organization is also supported by standards defining automation functions and methods for specific building infrastructure subsystems. All this constitutes a new opening of BIM application possibilities in the abovementioned design process but also in the effective and dynamic operation of modern buildings discussed in Section 3.

Another approach and technology that supports the monitoring and effective management of modern buildings is the DT. However, it should be noted that this approach is intended to be universal and can be used in many industries and application areas. As indicated in [40–42], DTs comprise a technology used in various sectors such as industry, engineering, healthcare, and others. Moreover, they are dynamic in nature, depending on the character of the objects for which a digital replica is created. The publication [43] establishes the idea of the DT. This shows how supply and demand not only for products but also for functionality will drive the market and the direction of technology development. Modern DTs can work with various types of data, static and dynamic (real-time), which enables them to perform interactive simulations, predict potential threats, develop used scenarios, and so forth. Consequently, Aheleroff et al. [27] identified three organizational areas of modern DT applications, as illustrated in Figure 2, which support the implementation of BACS systems, enhance the smartness of buildings, and facilitate collaboration between the physical and virtual layers.

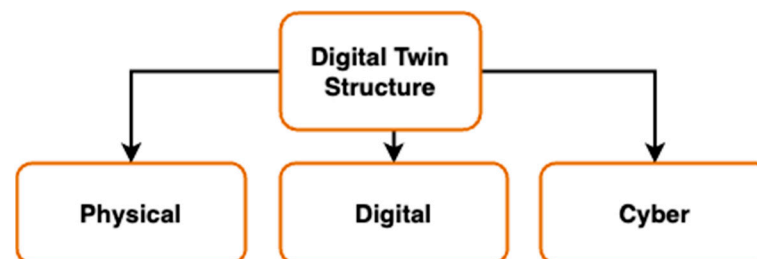


Figure 2. Digital twin basic composition.

The typical DT consists of three principal components, which correspond to a distinct stage or function [27,28]:

- A physical object is an actual object that is modeled by a DT. It can be any physical object, such as an entire city, in extreme examples. The physical object is equipped with every type of sensor and other devices that measure and record data, which are then transmitted to the digital components of the DT.
- A digital model is created using a variety of techniques, including 3D modeling, computer simulations, and ML. The digital model contains detailed information about the physical object, its parameters, current state, mode of operation, and interdependencies with the environment and users. DTs are classified into three categories: machine, product, and process [44]. The first of the digital machine twins are used to model and simulate machine operations, which enables the prediction of failures and the opti-

mization of maintenance. The second digital product twins facilitate product design and testing by digitally mapping them. Finally, the digital process twins facilitate the identification of areas for improvement both considering real data and predictions based on historical data.

- A cyberdata processing system combines a physical object with its digital model. This system is responsible, among other things, for collecting and storing sensor data, processing it, and updating the digital model in real time. The data processing system may also include ML algorithms that allow the prediction of physical object behavior and the identification of potential problems, diagnostics, and inspection planning [45].

The aforementioned elements render the DT approach particularly useful in supporting building and infrastructure management procedures during the building's operation period, complementing the solutions and possibilities offered by BIM [46]. It is of particular importance to note that DT solutions utilize real data from the serviced facility, which predisposes them to cooperate with extensive modules of modern BACS and smart home systems and also from the perspective of integrating IoT solutions.

A further advantage of both BIM and DT approaches is the availability and functional capabilities of utility tools and software packages supporting the organization and operation of BIM and DT tools. The availability of this type of tool package is significant, but in practice, in relation to the building industry, it focuses on several solutions.

CarnotUIBK version 2.1 is MATLAB's version R2024a new tool for dynamic building simulations that can be used for DTs. It consists of two parts: an object-oriented data management framework and a simulation framework in Simulink version R2024a. It allows one to easily import data from various formats, including gbXML version 7.0 (BIM), Excel version Microsoft 365, and PHP version 7.0. Moreover, this tool provides efficient simulation and result analysis, very important in verification procedures of various operating and control scenarios of building installations. It is adapted for "hardware-in-the-loop" applications, development of controllers, and internal air quality simulation. It is a valuable tool for civil engineers, HVAC system designers, and researchers [47].

Revit is a widely used BIM design software program for construction. According to [48], there is a clear need for training programs for technicians and engineers on the use of Revit in architectural design (Revit Architecture), installations (Revit MEP), and costing (Revit Costing). The research and training programs currently available at universities are inadequate. The implementation of training programs for midlevel staff can facilitate the utilization of BIM technology in construction.

In [49], the building management problem was solved via cloud service Autodesk BIM 360 Ops. This service enables the manager to add asset data from various sources such as Revit, BIM 360 Field, IoT, and spreadsheets. This facilitates review documents and models, maintenance checklists, schedules, and history. This approach enables maintenance workers to manage assets and perform maintenance tasks.

As indicated in [50], the Ansys Twin Builder software version 2024 R1 facilitates integration with SCADE Display visualization software version 2024 R1. This functionality enables Ansys users to observe movements, deformations, and other parameters of a simulated object in a 3D environment in real time. This program allows one to visualize complex simulations covering many areas of physics (e.g., mechanics, heat flows, fluid flows) and various engineering disciplines (e.g., mechanics, electronics, control). This approach enables engineers to have a holistic view of the simulation and identify the relationships between the various elements.

When selecting software for the creation or operation of a DT, it is essential to consider the user's specific, current needs and anticipate future needs. It is necessary to check which functions the software performs and whether it is intuitive to use, which may be interpreted differently by different individuals. A good example is CarnotUIBK, which is highly effective for energy simulation of buildings but has a more limited range of applications than other BIM tools such as Revit or BIM 360 Ops. Nevertheless, despite its wide range of building element modeling capabilities, Revit is not designed to create DTs

and requires additional software for simulation and analysis. Similarly, while Autodesk BIM 360 Ops is focused on construction project workflow and is great for managing data throughout an object's life cycle, it can be too complex and extensive for simple projects. The BIM software (<https://www.autodesk.com/solutions/aec/bim>) programs market is expected to grow rapidly in the coming years, so more solutions will emerge. It is expected that some will be developed with a specific focus on building automation, while others will have more general applications.

2.1. Development of the BIM and DT Applications—Key Challenges and Gaps

The actual and potential multitude of applications of BIM and DT tools in the construction industry and the organization of infrastructure in buildings and their surroundings generates a number of significant challenges. Moreover, an analysis of the literature and research results also indicates several gaps in these areas. They concern a number of issues related mainly to a new approach to the effective utilization of the aforementioned tools and techniques, which support the design procedures, integration, and increase in the accuracy of modeling operational processes. Furthermore, they encompass the consideration of innovative IoT networks and BACS functions related to the extensive infrastructure of modern, smarter buildings. Figure 3 provides a summary of the key categories of issues, challenges, and gaps discussed in this review paper.

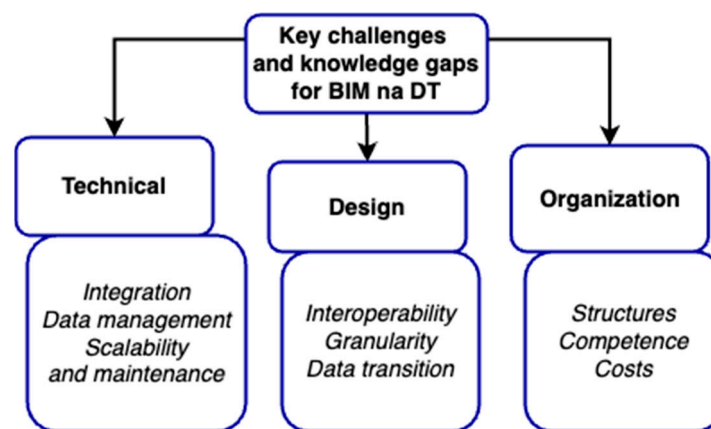


Figure 3. Key challenges and knowledge gaps for BIM and DT.

2.1.1. Technical Challenges and Gaps

One of the key areas where knowledge and good practices are still lacking is the integration of BIM and DT, as indicated in the papers [23,30,51,52]. According to them, to address this issue, it would be necessary to examine in this context the use of open data-sharing standards such as Industry Foundation Classes (IFCs) for BIM and cloud-based data sharing for DT. The authors conclude that all stakeholders should cooperate and establish clear rules on data sharing and management. For instance, BuildingSMART International is an organization that develops IFC standards for BIM. Standardization plays a key role in overcoming many of the challenges and obstacles associated with the integration of BIM and DT. Firstly, it would ensure data consistency and minimize the risk of information loss during transition between different project stages. Furthermore, it would enhance efficiency and facilitate collaboration and the flow of information, saving time and resources. All these issues are crucial in the context of the application of BIM and DT tools in the structures of distributed fieldbus automation networks, where network nodes are characterized by limited computing power and memory resources (edge modules) [11,53,54]. Among the various initiatives in this area, it is worth paying attention to the activities of the Digital Twin Consortium [55]. It develops frameworks and best practices for various DT applications, including standards for data exchange and sharing between different tools. Nevertheless, all the mentioned initiatives and research and development works are in the

implementation, experimental phase. Consequently, there is currently no single, universally accepted standard for BIM and DT data integration.

It is of paramount importance to ensure that real-time interaction between BIM models and DTs is maintained in order to guarantee the accuracy and consistency of data, as well as to facilitate seamless interaction between different project stakeholders [23]. The advantage of real-time interaction is enhanced communication, meaning instant information exchange between engineers, designers, and contractors. Access to up-to-date project data facilitates the ability to make quick and accurate decisions. Furthermore, it allows one to mitigate errors through express punching and problem solving even at the project stage as well as avoid costly alterations and delays in construction, so it also carries an economic aspect. All these issues are particularly important from the perspective of the use of BIM and, in particular, DTs in the organization of dynamic energy management platforms and energy media in buildings and related prosumer microgrids [56]. As emphasized by Bortolini et al. [25], the combination and analysis of real-time and historical data for specified building infrastructure allows one to conduct an effective strategy to protect, forecast, and optimize building energy efficiency. With this approach, the DT can accurately portray real-time operational circumstances and predict future states of building infrastructure elements based on continuously acquired sensing data and their relationship to historical data.

However, maintaining data consistency between BIM and DTs can be difficult, particularly for large and complex projects. Appropriate technological infrastructure such as a high-speed Internet connection and collaboration software is essential to ensure seamless interaction. Moreover, it is crucial to guarantee the security and confidentiality of data shared in real time. Development of a data management strategy that determines who has access to the data and how they are updated is paramount as well [28].

2.1.2. Design Challenges and Gaps

It is possible to encounter difficulties at any stage of the integration process. One challenge is the multitude of file formats and data exchange standards used in various BIM software programs and DT platforms. One of the key priorities in the development of tools to translate data between different formats is to ensure that this process is as efficient as possible. It is inevitable that information will be lost when data are transferred between stages [52].

Another problem is the detail and breadth of the projects. BIM and DT models that are too detailed can be too heavy and time-consuming to manage. Models that are too detailed may not provide sufficient information to enable decisions to be made. There are serious difficulties in adapting the level of granularity to the needs of a given stage of a project. One potential solution may be LOD (level of development) models, defining levels of detail for BIM models, as discussed in [56]. This concept is considered in the context of implementing the FM strategy in buildings and direct support for facility managers to address the challenges of information reliability, interoperability, and usability related to the service and operation of building infrastructure [57]. The advancement of this tool could also be considered and used in a DT environment.

In the context of designing building infrastructure and building automation systems supported by BIM and DT tools, it is important to consider the problem of fragmentation and granularity that arises from the technical nature of these systems [58]. Modern BACS are distributed in nature, and assuming their integration with IoT solutions, other fieldbus networks, and edge-level modules [2,10,11,54,59], the granularity will increase, which is a problem and a challenge in building effective BIM but also in the organization and integration of data acquisition structures for DT, as discussed in [60].

2.1.3. Organization Challenges and Gaps

Another knowledge gap is the realistic handling of a BIM-DT combined project. Traditional organizational structures may not be adapted to the cooperation and information flow required for the effective use of BIM and DTs. In the paper [28], Coupra et al. iden-

tify several significant gaps and related challenges in the implementation of BIM and DT methods and tools to support the effective operation of smart buildings. These include the following: (i) the lack of standards to unify and integrate tool platforms and data processing; (ii) the lack of up-to-date data resulting from a lack of communication between various stakeholders, controllers, and systems supporting various elements of building infrastructure; and (iii) the lack of organizational strategies that would take into account all stakeholders, in particular those emerging on the horizon as a result of new requirements and needs of contemporary buildings and their users.

As emphasized by Yang et al. [30], there is still a lack of a sufficient number of diverse case studies analyzing the conditions and application possibilities of BIM and DT in the organization of BACS systems, including the integration of IoT on various facilities. This is particularly important in the context of the implementation of BIM and DT methods in new areas related to FM and the organization of mechanisms and systems supporting the improvement of the energy efficiency of buildings. Furthermore, the implementation and maintenance of BIM and DT platforms, in particular integrated ones, involves additional costs that companies as well as investors must consider. Prior to the implementation of BIM and DT, a meticulous cost–benefit analysis is essential to ensure that the investment yields a positive return on investment [26,61].

3. BIM and DTs—Latest Development Trends and Challenges

Bearing in mind the most important features, application predispositions, and gaps identified for the BIM and DT approaches in Section 2, there are many publications in the literature that discuss and analyze their development trends. Considering the subject matter of this paper, the focus is on trends and challenges related to the intelligent building industry, which supports advanced system infrastructure and automation. The advent of these advancements has led to an increasing demand for BACS functions dedicated to supporting various building infrastructure subsystems, along with mechanisms for their effective integration. In order to address this challenge, the concept of automation functions is defined as services to be performed by distributed network systems, including IoT networks, utilizing cloud resources and services. This approach has been proposed and defined in the technical report [19] related to the SRI definition and evaluation. However, considering the potential for realizing automation functions in distributed IoT networks in the papers [54,62,63], the authors propose the concept of control services as a universal organizational element for data objects and their functional integration. Furthermore, based on this approach, concepts for the organization of buildings as services (BaaS) are being formulated. These concepts allow for the dynamic shaping of functional frameworks, which can be adapted to changing user needs and the utility nature of such facilities [5,63–66].

3.1. Designing, Modeling, and Control as Services

The dynamic shaping of the usable and functional environment of buildings requires a change in approach to the organization of BMS systems with FM elements, in particular, the implementation of DT structures as technical solutions that facilitate the effective development of building and BACS functions and services. One of the unobvious trends emerging in the market is the concept of the digital twin as a service (DTaaS), which refers to the use of DT models in the cloud. The idea is that customers will use an online platform, paying only for the services they use without having to maintain their own infrastructure. The cloud platform on which the DT tools are stored is made available to users. The data are transmitted from BACS devices, BIM models, and IoT sensors. The platform is responsible for integrating the data and creating a real-object model for customer use. The DTaaS offers a range of applications, from design and construction to industry. The tool is beneficial in the management of buildings, monitoring user comfort and parameters such as energy consumption [28]. Furthermore, the application can also be useful during the ordinary maintenance of a building, supporting and communicating the need for maintenance, anticipating failures, and planning repairs [26,27,55,67–69].

One of the most significant benefits of using the DTaaS is the potential for saving time, resources, and money. It has much lower initial costs by not needing to invest in servers and software. The customer is not required to possess the specialist knowledge typically provided by the service provider and platform [24,59,70]. Moreover, according to Bortolini et al. [25], companies using these services definitely have a competitive advantage through better management and monitoring of their facilities. Such solutions can be tested in a virtual environment, aiming to accelerate innovation. By optimizing resources and energy consumption, they remain “green companies” and reduce their environmental impact. Considering the mentioned BaaS frameworks, the DTaaS approach also aligns with the development trend of virtualization of BMS and FM platforms (emerging issues in the BMS “cloudification” endeavor) and transferring the functions of collecting, processing, and analyzing data from distributed automation and IoT networks to cloud services [71–73].

It is important to note that the exponential growth of ICT technology has a profound impact on the evolution trends of modern network automation systems in both the industrial and building sectors. In recent years, there has been a fruitful convergence of different technologies, microelectronics, and building systems towards an IP-based infrastructure supported by proprietary communication protocols as well as ethernet and Internet networks. Technological convergence in the management of buildings and smart buildings is accelerating with the increasing deployment of IP-based devices within the IoT. According to [74], building systems used and still are using various protocols, networks, and cable systems. This is evidently an inefficient approach from both an implementation and a management perspective. The IoT concept, which is now being used more and more often, will further enhance, standardize, and expand the functions and scope of BACS services. BMS systems are systematically and consistently switched to IP networks (which also allow remote monitoring of many buildings through a central operations center). The growth of the IoT is being driven by a sharp decline in the cost of advanced sensors, computing power, and data storage along with increasing device density (and decreasing device size) [11,72]. The IoT can support BIM and DTs by collecting building status data, monitoring performance, predicting failures, automating tasks, and personalizing the environment. Combined with BIM and DTs, the IoT can create smarter, more efficient, and more sustainable buildings [32,75]. In evaluating the potential of BIM methods and tools for this type of application, it is also essential to consider their internal development trends. BIM goes beyond just 3D modeling, now offering tools for construction schedule planning and control (BIM 4D) as well as cost estimation (BIM 5D) [23,24,30]. BIM software is evolving towards a comprehensive solution, encompassing tools for the design of structures and plumbing, electrical, and HVAC systems [28].

3.2. Implementation of IoT Paradigm and Data-Based Solutions

The basis for each of the aforementioned solutions is data, resulting from both design assumptions and standard requirements, as well as operational data. This includes information on the operating status of building infrastructure subsystems, the intensity of the utilization of rooms, and the environmental conditions prevailing there. Moreover, there have been significant changes in the manner in which these data are acquired and processed for the purpose of utilizing BIM and DT models in the context of building applications. Now, instead of relying solely on BIM data, engineers and integrators combine them with information from fully distributed IoT sensors, monitoring systems, and other sources. This results in a more comprehensive DT that more accurately reflects the real object and is directly related to BACS and BMS functions. The trend is to enrich the DT with data from different sources (BACS sensors and actuators, smart meters, etc.), allowing for better modeling and analysis of reality and facilities as a whole [23,76–78]. Furthermore, instead of relying just on traditional methods (periodic monitoring, measurements), with DTs, engineers are able to create realistic simulations. These virtual experiments predict the future functioning of a building, thereby optimizing its performance in a multitude of ways. In addition, they enable testing of new technologies, such as photovoltaic panels or intelligent

energy management systems, in a secure virtual environment [47,79,80]. Moreover, these virtual copies of buildings can not only simulate how they work but also assess the impact on the environment. Engineers can test different building materials, heating systems, or ventilation solutions in the DT to choose the most environmentally friendly ones. In this way, DTs make it possible to design energy-efficient buildings that use less water and emit less pollution [25,79,81].

Making full use of the potential of BIM and DTs in the construction industry faces a number of technological and organizational challenges. In [52], Vieira et al. discuss three main data quality factors, completeness, complexity, and flexibility, in the context of organizing advanced BIM for various BACS open standards: KNX [82], LonWorks [83], and BACnet [84]. Bearing in mind all elements of BACS standardization like communication media with network topology, network variables, and data objects, as well as functional profiles with logical bindings, the authors analyzed usefulness of the BIM concept in advanced BACS and BMS platform design and integration. Additionally, the paper [30] provides a list of the most important research trends/challenges for BIM developments in the context of BACS design and integration support. These include the following: (i) the establishment of an open platform for the integration of multiple BACS and smart building technologies, (ii) the multidimensional consideration of BIM software standards, and (iii) data format interoperability with universal data tagging.

Considering these trends, it should be noted that in the area of automation, the lack of system integration, the variety of data formats, and the shortage of qualified staff with the right knowledge and skills represent the most significant obstacles to be overcome in the field of automation. Project challenges are focused on the problems of data exchange between BIM/DT platforms, the lack of seamless data transition between project stages (from design to operation), and the need to optimize the level of detail of models according to the specificity of the stage. Optimizing organizational structures for efficient BIM/DT use is also a key aspect, according to [23]. Once these challenges have been overcome, BIM and DTs have the potential to become powerful tools that can significantly increase the efficiency of construction projects, reduce costs, minimize errors, and improve the overall profitability of projects. Therefore, as indicated in [85], BIM and the IoT promise a continuous flow of information, but the connectivity between model and reality can be incomplete (e.g., software changes, unlisted parts, software version errors). These drawbacks are also reflected in the world of DTs.

In order to enhance the degree of interoperability and integration of system data, a number of engineering and research initiatives have been implemented. The primary objective of these efforts is the establishment of universal standards for communication and data identification. In the case of BACS, standardization of automation functions and services is also a key objective. This included the implementation of and adherence to common standards for data exchange and development of a data management strategy that ensures consistency, security, and availability throughout the project life cycle. The natural direction is to develop open and accessible BIM and DT platforms, as well as to develop and implement standards with best practices for their use [23,54]. It is necessary to ensure adequate funding for the implementation of BIM/DT platforms and to invest in the training and education of employees to increase the number of qualified staff. Moreover, it is crucial to implement effective management of organizational changes to facilitate the transition to new working methods and to promote the culture of open collaboration as well as information sharing between all project participants.

One of the most rapidly developing trends in the context of universalization of communication and data for the integration of BACS, BIM, and DTs is the implementation of IoT solutions. According to [86], it is critical to understand this still emerging paradigm to address the multifaceted challenges associated with the large-scale deployment of DTs within industrial and building IoT applications. Moreover, ref. [87] analyzed both the workflow between BIM and the IoT and the resulting cloud-based DT in this combination, along with functionalities that were still in testing. One of the most significant challenges

associated with this process is the security of data and the reliability of communication, particularly when utilizing a radio communication medium such as Wi-Fi. As discussed in [75,88], the challenges of IoT security include, among other things, the inherent problems of IoT security. There are inherent security vulnerabilities in IoT devices that make it difficult to protect them. In addition, use of vulnerable Internet gateways/concentration points. Gateways and concentration points that collect data from multiple devices can be an attractive target for cyberattacks. Low-complexity devices and simple IoT devices often have limited security options. The open nature of the environment allows for physical access and manipulation by hackers. The limited power supply of low-power devices precludes the implementation of more complex and effective safety algorithms. It is important to note that IoT devices can be portable and difficult to track, making it difficult to ensure security. Moreover, they are often constantly connected to the network, increasing the risk of being attacked. Additionally, the main advantage of the IoT approach is the vast number and variety of IoT devices, which makes it difficult to develop universal security solutions. The lack of a unified IoT security architecture makes communication and data protection difficult [75,89,90].

Finally, it is worth mentioning one of the latest trends in the development of many industries—the idea of using artificial intelligence (AI). It is one of the common trends, and not just for industrial and building automation. AI is entering BIM with momentum, offering a range of facilities for designers and engineers. It is a valuable tool for automating tedious and repetitive tasks such as creating documentation, clash checking, and quality control, allowing one to save time, resources, and money. Furthermore, AI can be employed in the analysis of complex BIM data, suggesting changes to improve performance such as reducing energy consumption. In addition, it generates photorealistic images and 3D videos, thereby facilitating the visualization of a project [91–93]. One of the advantages of AI in BIM is that it costs less to detect errors early and optimize projects at each stage, generating savings. Such technologies facilitate creativity by giving more time to people. And at the same time, they help to create better buildings and aesthetic designs, supporting sustainable development. The AI trend in BIM is still in its nascent stages of development, yet its potential is considerable. It is advisable to monitor the development of this technology in order to gain a competitive advantage and to undertake projects at a completely new level [57,94,95].

4. BIM and DTs as Tools to Support BACS Design and Management Processes—Opportunities, Challenges, and Research Directions

Considering the original contributions defined in Section 1 and the trends and challenges discussed in Section 3, the authors of this review propose expanding the concept of utilizing BIM and DT methods and tools in the domain of the design, construction, and operation of buildings. In particular, they should consider two main development trends in building innovations. The first is the increasing smartness and comfort of use, and the second is the improvement of energy efficiency, the utilization of RES, and the readiness for smart grid solutions. In light of the information presented in [26,27,69], it is evident that an important element of this concept is the required integration of the digital and physical layers. Therefore, the authors of this paper propose cloud services as an integration platform also for the implementation of BIM and DT services, as shown in Figure 4.

In the context of designing BACS and BMS functions based on fieldbus and IoT network modules, it is essential to refer in BIM to functional blocks and network variables related to specific modules of sensors, actors, and available cloud services within the BIM framework. This enables the implementation of more advanced functions and data analytics. Consequently, at the stage of construction and operation of these systems within the building, hardware and software integration is necessary, allowing the acquisition and use of data from these modules in DT models and mechanisms. This enables the development of optimal operating parameters for building infrastructure devices and the effective implementation of predictive models. At the same time, such DT models

can facilitate mechanisms for enhancing the energy efficiency of a building thanks to the selection of parameters of the BACS and TBM functions, not only during the design phase but also during the period of dynamic operational changes.

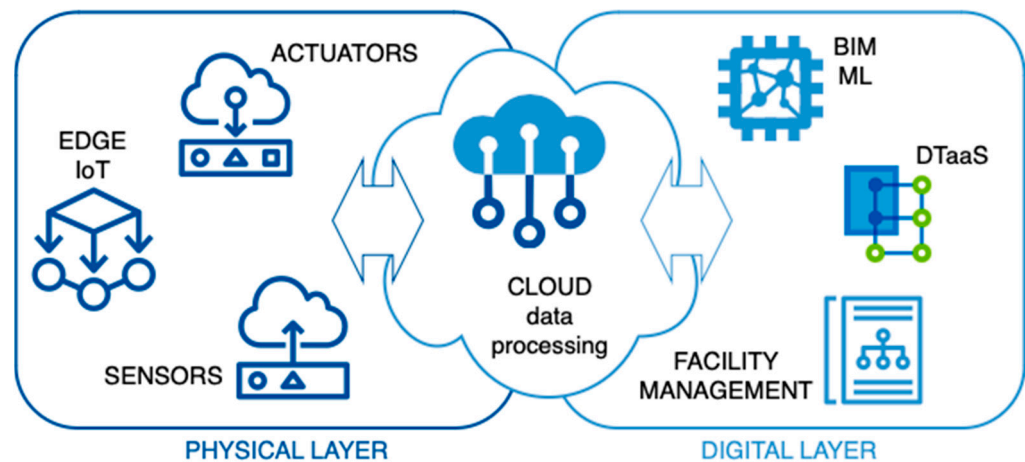


Figure 4. The concept of an integrated BIM and DT platform with BACS and IoT support in buildings [26,27,69].

4.1. Standards, Requirements, and Approaches

A novel approach is required, particularly in light of the standards and guidance documents published in recent years that regulate the possibilities and rules for selecting BACS functions and smart services dedicated to buildings. One of the significant and widely applicable standards regulating the principles of utilizing BIM in the organization of building functionality is the ISO 19650:2018 “Organization and digitization of information about buildings and civil engineering works, including building information modelling” [96]. It provides a set of guidelines for the management of a building throughout its life cycle. Moreover, the document highlights the potential of BIM as a valuable tool to enhance the utilization of facilities, the operational efficiency of buildings, and the effectiveness of design processes. This standard contains several sheets covering detailed issues: (i) the flexibility and versatility that characterize the wide range of potential BIM strategies, (ii) the requirements for information management in the context of the delivery phase of assets and the exchange of information within it using BIM, (iii) information management in the form of a management process in the context of the operational phase of assets using BIM, (iv) the detailed process and criteria for decision making when performing an information exchange as specified in the ISO 19650 series to ensure the quality of the resulting project information model or asset information model, and finally, (v) the principles and requirements for security-conscious information management for BIM as well as for the security-conscious management of sensitive information that is acquired, created, processed, and stored as part of any other initiative, project, asset, product, or service. As the analysis of these issues demonstrates, the implementation of BIM in modern buildings requires advanced technological tools—not only software but also hardware. Consequently, the authors of the concept of a seamless marriage of BACS technologies, IoT, and cloud services propose that this could be a potential platform for the effective design of BACS functions and extended automation networks in buildings, as well as for the organization of FM tasks during its operation [97].

As an illustration of the initiatives that have already been implemented in this field, in particular aimed at the universalization and unification of information about buildings for BIM systems, it is possible to provide the development of an open standard for formatting COBie (Construction Operations Building information exchange) information [98,99]. It is a key part of the digital transformation of construction, which enhances building management and reduces costs. This open standard provides a set of rules for universal data exchange about the building that underpins the DT. It simplifies data exchange between

systems such as BIM, CMMS (computerized maintenance management system), and BMS. Moreover, it ensures a good level of data quality and consistency while facilitating a faster response during a crash.

However, it should be noted that in the case of BACS, in particular the open standards of KNX, LonWorks, and BACnet building automation, standard data objects that are commonly used in the industry are already defined (e.g., SNVT—standard network variable types for LonWorks and DPT—data point types for KNX). Moreover, functional profiles have been defined for these standards, which determine the principles of implementing the most important automation functions, using standardized data types. Therefore, when trying to integrate the BIM and DT environments as tools for the design and maintenance of BACS and BMS, it is essential to consider these standards, along with their potential for developing new, universal semantics. An example is the Haystack project [100], with the idea of standardizing semantic data models and web services with the goal of making it easier to unlock value from the vast quantity of data being generated by the smart devices that permeate homes, buildings, factories, and cities. The fundamental elements of the semantics are tags defined for various kinds of data from BACS and BMS applications including automation, control, energy, HVAC, lighting, and other systems.

4.1.1. BACS and Energy Efficiency Performance

The rapid development of BACS technology over the past two decades has resulted in the need to systematize not only communication and data transmission standards but also the automation functions themselves along with the principles of their functional integration. In the first decade of the 2000s, the EN 15232 standard entitled “Energy performance of buildings—Impact of Building Automation, Controls and Building Management” was developed, which was updated several times. The latest version from 2017 [14] introduces four categories of BACS, marked with letters from A to D depending on their impact on a building’s energy efficiency, as shown in Figure 5 [8,101]. It is crucial to note that these categories differ from the energy efficiency classes of entire buildings as defined in the EN ISO 52003 standard [102].

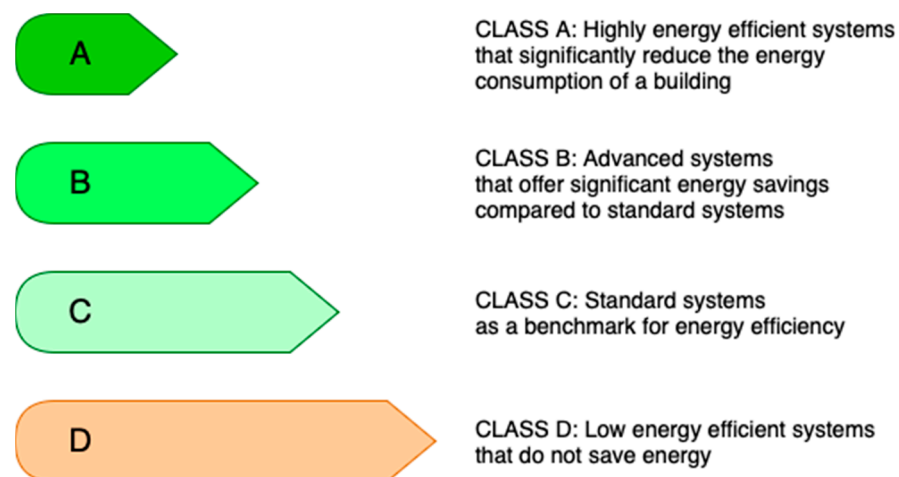


Figure 5. Categories of building automation systems from EN 15232:2017.

As shown in [103], the standard can be used already at the design and prototype stage. The simplified list of control functions with a basic indicator calculation method could be used for buildings without detailed information related to BACS and TBM concepts. This allows for the optimization and evaluation of different design and functional variants. At the construction stage, the building can be used to identify areas where energy consumption can be additionally reduced. On this basis, integrators could consider various variants of temperature and lighting, considering daylight availability, occupancy factors, etc. However, as indicated in [104], there is a gap between the assumptions of EN 15232:2017

and the actual performance of BACS. The standard suggests that the higher the level of automation, the greater the energy savings, but the practice shows that users can circumvent these systems to meet their needs, for example, by blocking light or presence sensors [5,105]. Users may not be aware of the impact of their actions on energy consumption. Therefore, it is important to raise awareness and educate both professionals and ordinary household members. Moreover, case studies for different kinds of buildings are very important and useful, considering not only various use scenarios but also different geographical locations of buildings. This represents a clear application area for BIM during the design and construction phases, as well as for DTs during the operation of buildings with BACS and BMS.

In 2021, the EN 15232 standard was replaced by the EN ISO 52120 standard [15], which introduced minor changes and sanctioned the significance of the TBM function in BACS systems and advanced BMS scenarios, particularly in the context of active monitoring and management platforms of local energy networks—microgrids. The EN ISO 52120 standard applies to the energy performance of buildings with the BACS and TBM functions. According to [106,107], it raises awareness of the benefits and requirements of building automation for energy efficiency in buildings. Moreover, it defines building readiness levels for integration with smart grid systems and provides methods for verifying the potential of BACS functions and installations in the area of improving the energy efficiency of buildings, as illustrated in Figure 6.

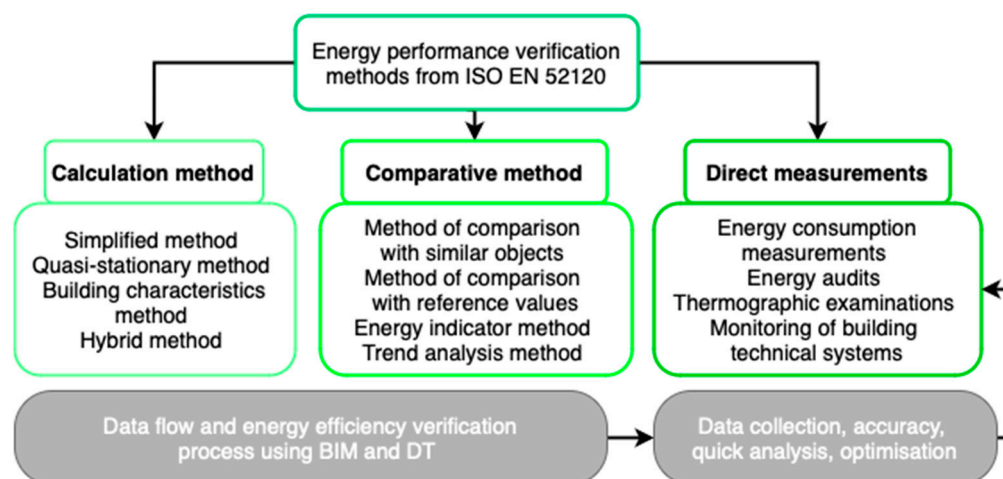


Figure 6. Energy performance verification methods from EN ISO 52120 with potential BIM and DT application support.

The standard defines and describes three verification methods, along with performance indicators, for the calculation and comparison methods. All of these methods have the potential to utilize BIM and DT tools in both the design and verification procedures of BACS and BMS installations. Although the standard considers a multitude of factors affecting the energy efficiency of buildings, there are some key elements that are overlooked. The most crucial aspects are stated by Vandenbogaerde et al. in their paper [13]. The system presented by the authors considers and analyzes the load on the building, user behavior, climatic conditions, and maintained temperatures, adapting the operation of the systems to adjust energy consumption to these parameters. Furthermore, the authors consider technical parameters such as control algorithms and sensor resolution to ensure precise control and optimization of the building operation. The location of the building is analyzed in order to optimize the use of sunlight and reduce the need for artificial lighting. Moreover, the system considers the ratio of window surface to wall surface, window–wall ratio (WWC), to ensure the right balance between natural light supply and thermal insulation. According to authors, by taking these factors into account, the BIM/DT system can significantly reduce the energy consumption associated with lighting and heating/cooling a building.

The omission of these elements may result in an underestimation or overestimation of the energy saving potential of the building.

DT technology was also analyzed in the publication [108]. It was pointed out that DTs are a widely analyzed subject in a large number of citations also in relation to thermal comfort. Regions of the world were indicated where knowledge in this area should be better explored. Critical analysis presented factors that are reproducible in many publications, such as humidity and air temperature and the direction of future analyses in terms of factors such as air velocity or mean radiant temperature.

4.1.2. BACS and Smartness of Buildings

Based on experience from the implementation of the EN 15232 and EN ISO 52120 standards, the Smart Readiness Indicator (SRI) was launched in 2018. Initially, the SRI was incorporated into the Directive of the European Parliament and the Council of the European EPBD 2018. Since then, further work has been conducted, which concluded in 2020, resulting in a detailed methodology for evaluating the readiness of buildings for intelligent solutions, discussed in detail in the technical report [19]. Voluntary implementation by EU countries is currently underway. Importantly, the SRI assesses the readiness of buildings to be “smart” rather than their current performance or energy efficiency [16,22,80,109–112]. However, the aforementioned report contains a crucial element: two lists of functions and services of building automation systems, dedicated to various building infrastructure subsystems (domains). For each individual function, several functional levels are defined and described, relating to the level of advancement of the automation functions—smart services. The lists of functions are based on the lists of control functions described in the EN 15232 and EN ISO 52120 standards but have been extended to include functions and services dedicated to the operation of energy management systems. These include, for example, the charging of electric vehicles, the monitoring of energy consumption and power quality, the integration of renewable energy sources and energy storage, and the support for demand-side management and demand-side response mechanisms. Consequently, these lists can be employed as a framework for the design of BACS and BMS systems in accordance with the principles of BIM and DT tools. Furthermore, in a manner analogous to the EN ISO 52120 standard, the report delineates three methodologies for calculating the SRI value, presented in Figure 7. The initial two methodologies are based directly on the functional analysis and selection of automation functions/services, while the third methodology is the most precise and necessitates the measurement and analysis of the system already operating within the facility. In this context, it is possible to utilize DT tools as a source of data feedback for the calibration and verification of BACS control parameters.

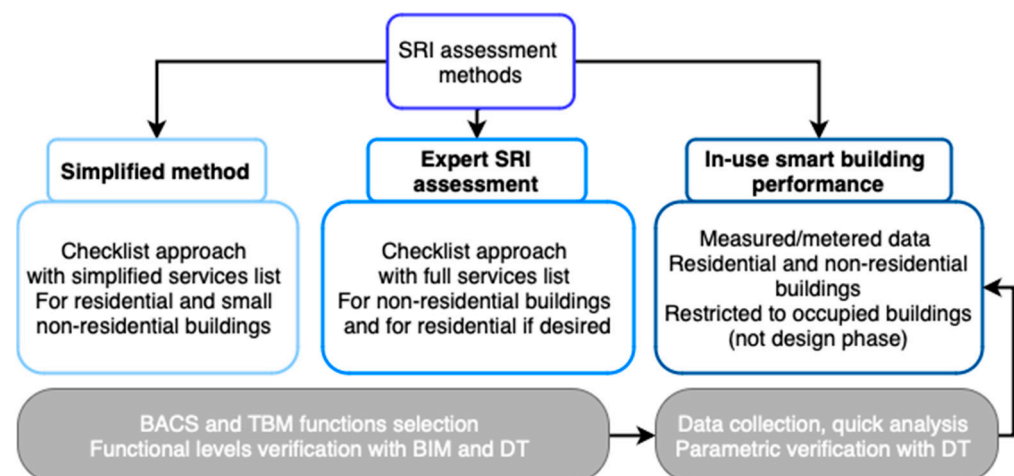


Figure 7. Three assessment methods for SRI with potential applications for BIM and DT.

The accurate and precise evaluation of the SRI allows one to identify buildings that may contribute to energy production. This is important in the context of urban management. Furthermore, the SRI would be beneficial in the management of power networks in terms of generation, transmission, and distribution. In this context, it is particularly important to consider the role of public buildings, such as educational institutions.

Bearing in mind the results of the study presented by Plienaitis et al. [109], and combining them with the guidelines for BACS systems and their effects of improving the energy efficiency and smartness of buildings, the energy efficiency of buildings is inextricably linked to the level of building automation and control systems for the technical systems of a building. The aforementioned study demonstrates the need for parallel and simultaneous consideration of both assessments in order to obtain a more comprehensive assessment of the overall energy performance of the building. Furthermore, the study [113] proposes that the SRI level of a building should be considered in the EPC (Energy Performance Certificate) evaluation. In the future, energy audits could also encompass “smart” buildings, and activities related to the modernization of energy systems could be adapted to take account of the modernization of smart systems. Further integration of the SRI into the energy efficiency assessment of buildings is necessary [114]. As illustrated in Figures 6 and 7, the joint application of DTs and BIM can significantly accelerate data collection and analysis, thereby increasing the accuracy of computational methods. This will facilitate the optimization of the design and continuous monitoring and improvement of the SRI level throughout the building life cycle.

4.2. Perspective for New Solutions

The DT and SRI can be used to facilitate more informed decision making given that both are dynamic and possess the capacity to learn. They have the potential to collaborate on various stages of a project. The DT enables the simulation of scenarios, while the SRI indicates the optimal actions to be taken, thereby contributing to the optimization process. The DT can be updated in real time as new data arrive, while the SRI can adjust its metrics based on changing needs and conditions. The common features of the DT and SRI provide a starting point for further research and development in the area of intelligent systems. The integration of these two novel systems opens up new possibilities and opportunities in areas such as modeling and simulation, as discussed in [115]. Together, they can contribute to the optimization of operational and building management. These tools are highly promising in the context of smarter buildings and cities [17].

At this stage, the SRI focuses on the technical elements of the building. Nevertheless, it is recommended that it should be extended with additional factors relegated to energy efficiency, user comfort, and sustainability. Indicators such as energy consumption, CO₂ emissions, user satisfaction, and the share of renewable energy sources should be included in the assessment of buildings. This approach, which considers the mentioned parameters, should be relatively easy to use in BIM solutions, facilitating the organization of more integrated BACS and BMS with developed IoT modules. Furthermore, at the design stage, it will facilitate more accurate parameterization of BACS functions, thereby providing a foundation for verification methods with DT applications.

In order to remain relevant in the context of the construction industry, both the SRI and EN ISO 52120 must keep pace with the latest technological developments and support the construction of energy-efficient and sustainable buildings in the digital age. The norm is currently lagging behind new developments in construction, such as BIM and AI. It is necessary to implement an update that considers the impact of these developments on data formatting and coding, which can be a long process. In essence, the objective is to stimulate and facilitate the utilization of digital tools for the assessment of the energy efficiency of buildings already at the design and construction stages. The standards should support open BACS and IoT standards and data formats. With this approach, different programs can communicate with each other freely and, at the same time, facilitate the creation of

digital platforms where data on the comfort management, security, and energy efficiency of buildings can be shared.

4.3. Important Challenges and Research Directions

The challenges and development trends of BIM and DT methods and tools were discussed in Section 3. However, considering the conclusions of the discussion in Section 4, the authors identify the most significant challenges and trends in the development of BIM and DTs. These are of particular importance in the context of supporting the implementation of BACS function design mechanisms and the technical organization of distributed networks with the IoT as well as their primary and functional verification at the stages of construction and operation in buildings.

As previously stated, the paramount concern in facilitating comprehensive and multi-faceted integration of BIM and DT software (<https://www.dt-software.co.uk/>) platforms with physical objects of distributed BACS networks with IoT guarantee the integrity and dependability of their data, as well as the reliability of their communication and processing. The availability of real data is the most significant factor in determining the efficacy of the utilization of BIM and DT tools in combination with BACS and BMS technologies, during both the design and operational phases. The EN ISO 52120 standard and the SRI methodology provide guidelines for the selection of functions, control services, and monitoring of buildings but do not contain any indications regarding preferred technologies. Therefore, an open environment is created for the implementation of various technical solutions in the field of distributed automation networks and the IoT. Nevertheless, as evidenced by the experience of numerous years of implementing BACS and BMS systems shows, the optimal outcomes are achieved by implementing installations based on open, nonproprietary standards [116–118]. The authors point out the necessity to develop novel mechanisms for the interoperability of communication protocols and communication media for object-level networks, with the objective of enabling data communication with higher-level networks (the Internet) while ensuring a high level of communication security, data coding, and the maintenance of the requirements for the operation of automation modules in the real-time regime.

Systematized guidelines for the selection and organization of BACS control functions and services specified in the EN ISO 52120 standard and for SRI represent an optimal set of parameters for implementation in BIM and DT tools. As previously stated, the implementation of these guidelines will not only facilitate the design and operation procedures of buildings but also allow for the benchmarking and performance tracking of such buildings. In particular, the SRI provides a standardized framework for the benchmarking and tracking of the building sustainability index over time. This enables building owners and operators to evaluate the efficacy of BACS and IoT-based initiatives and identify areas for further enhancement. Further support in this area will be provided by the integration of real system installations with DT models, which also enable the prediction of potential control scenarios and the analysis of dynamic changes in building utility parameters. These include variable energy tariffs, changes in load levels of household appliances, and the infrastructure of local microgrid networks.

Finally, it is crucial to emphasize the importance of proactive promotion of sustainable practices. This is particularly pertinent in the context of the implementation of effective energy management mechanisms at various levels, including the production and distribution of energy, as well as servicing of prosumers at the level of building complexes or individual facilities. Both EN ISO 52120 and SRI promote the implementation of sustainable building management practices by encouraging the integration of IoT technologies and cloud services to optimize energy efficiency, reduce environmental impact, and improve user comfort. Consequently, the implementation of these guidelines in BIM tools, which are becoming increasingly prevalent in the design and construction of buildings, will facilitate the dissemination and raise awareness among designers and investors of the implementation of sustainable development mechanisms at the building level.

5. Conclusions

This paper presents an overview of current knowledge on the development of BIM and DT techniques and their application areas. The collected information was analyzed in the context of the possibilities of its potential application in the technologically and functionally developing field of networked building automation systems integrated with IoT solutions. In this review, the authors discuss the directions of development of these solutions and models, as well as presenting their new trends and possibilities. In accordance with the preliminary diagnosis, assumptions, and objectives for this review, as outlined in Section 1, the analysis identifies gaps and potential barriers and obstacles to the advancement of the concept of synergies between BIM and DT methodologies in the design, construction, and operation of buildings with innovative BACS functions and services. These functions and services are oriented towards enhancing building comfort, improving energy efficiency, and facilitating the implementation of novel dynamic energy and infrastructure management mechanisms. As these processes are based on standardized guidelines (EN ISO 52120 and SRI), it is possible to describe them in BIM models as well as DT algorithms and patterns and, as a result, realize them using advanced tools for modeling, prediction, monitoring, and effective control of automation in buildings. As the authors note, this is of particular importance at the current time, when building infrastructure is becoming increasingly complex and multisystem. Furthermore, buildings equipped with RES and energy storage are becoming active participants in smart grid systems and the energy market as prosumers. This necessitates a shift in the methodology employed for the design, construction, and operation of buildings, particularly those equipped with BACS and BMS with IoT technologies. This is to develop universal and useful tools to support designers, installers, users, and maintainers. In this context, the authors formulated the theses of the original contribution concerning the possibilities of effective and constructive use of the EN ISO 52120 standard guidelines and the SRI, which can be effectively and constructively employed as integral parts of BIM and DTs in for contemporary buildings, both newly constructed and retrofitted. To meet the new standards, the market requires tools that accelerate work, thereby enabling the formulation of intelligent functions. An additional aspect is their operation and monitoring of parameters at every stage of the building's existence. It is for these reasons that the authors focus on a tool that could revolutionize the market and streamline many existing buildings, the DT. In particular, this concerns the technical and functional organization of BACS with distributed IoT networks, supported by cloud services for more accurate models, DT patterns, and parametric settings for control and management functions. Finally, this paper develops a conceptual framework for the application of BIM and DTs in these areas, identifies challenges and necessary research directions, and outlines perspectives for new technical and tool solutions.

Future Research Objectives and Work

In light of the findings and preliminary conclusions presented in Sections 2–4, and considering the challenges associated with the potential expansion of BIM and DT tools for the technical and functional design of BACS and BMS systems in buildings, the authors propose a number of key research objectives and directions for future research. These focus primarily on overcoming the barriers and gaps identified in this review. Here is their synthetic summary:

- A multitude of communication and data processing protocols have been developed for both categories of tools, with the majority of these protocols being developed independently by different entities. The effective cooperation of BIM and DT tools is necessary for the implementation of advanced functions and services of BACS and BMS systems. In order to achieve this, it is essential to reduce the number of protocols currently in use. Furthermore, the development of uniform rules for the representation of system data, which consider open building automation standards [74,119] (KNX, LonWorks, BACnet), would be beneficial;

- The objective is to develop mechanisms for the coexistence of communication protocols of the ICT network standard (TCP/IP) and communication in fieldbus networks, edge level, direct communication of sensors, and executive elements. In particular, the requirements for real-time data transmission for the implementation of active monitoring and control functions must be determined. This is to be performed in the context of implementing dynamic models of control and decision making in energy management systems, handling dynamic energy tariffs, and the rational use of resources in buildings and energy microgrids;
- The development of universal frameworks for the organization of standard monitoring and control functions for BIM and DT tools is required. These frameworks should be modeled on the functional profiles of open building automation standards and should consider the spread of fieldbus network nodes and the need for integration and interoperability of BACS, BMS, and ICT systems;
- The development of application strategies for BIM and DT tools in the novel area of building control and management, in particular in the broader perspective of improving their energy efficiency and facility management. Furthermore, it is necessary to raise awareness among building managers and their users regarding the benefits of implementing BACS and BMS systems with BIM and DTs, particularly in commercial, public, and industrial (nonresidential) buildings;
- The development and verification of mechanisms for data transmission and processing for BACS system platforms with BIM and DTs in the structures of distributed IoT networks and cloud services, with the development of real-time communication mechanisms between fieldbus network levels (edge nodes) and data servers in the cloud. Such solutions are necessary to support advanced functional models of BACS systems, with data analysis mechanisms in DTs for the rationalization of use scenarios and prediction of operating states, potential failures, damages, etc.;
- The opening of BIM and DT tool platforms to innovative learning models, including ML, reinforcement learning, and AI algorithms, represents a significant step forward in the field of construction. These novel solutions should facilitate the implementation of BACS and BMS functional organization procedures, in particular the selection of control and monitoring functions, their operating parameters, and the organization of effective integration algorithms for automation functions supporting various elements of building infrastructure and microgrid networks.

In consideration of the original contributions identified in this paper and the new research directions indicated in Section 4, future work of the authors of this article will concentrate on a detailed analysis of the BIM tools and design-simulation environments that are already available, with a view to determining their suitability as platforms for parametric verification of system designs and predictive analyses of automation and monitoring functions of equipment and building infrastructure. A further aspect to be considered is the necessity to ascertain the viability of integrating and utilizing DT models in the conceptualization and effective implementation of dynamic building management mechanisms. In their research and development work, the authors will focus on the development of concepts and technical solutions to automate the processes of selection and implementation of control, monitoring, and security functions of buildings with BIM and DTs in order to improve their energy efficiency while complying with the guidelines of the EN ISO 52120 and SRI standards.

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